

CUHK Scientific Layout Families

Model, evidence, imaging, and next-experiment pages

Research Presentation Program

Department of Statistics & Data Science

August 25, 2026



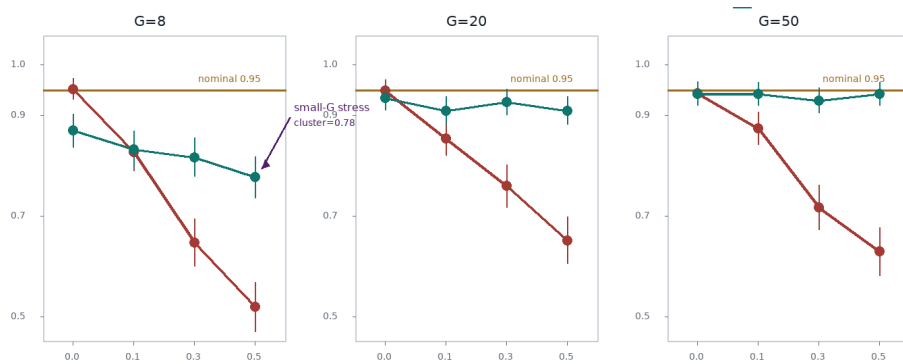
香港中文大學
The Chinese University of Hong Kong

Clustered outcomes separate center and individual variation

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}, \quad u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

Center effects set $\rho = \tau^2 / (\tau^2 + \sigma^2)$, so interval calibration depends on cluster count.

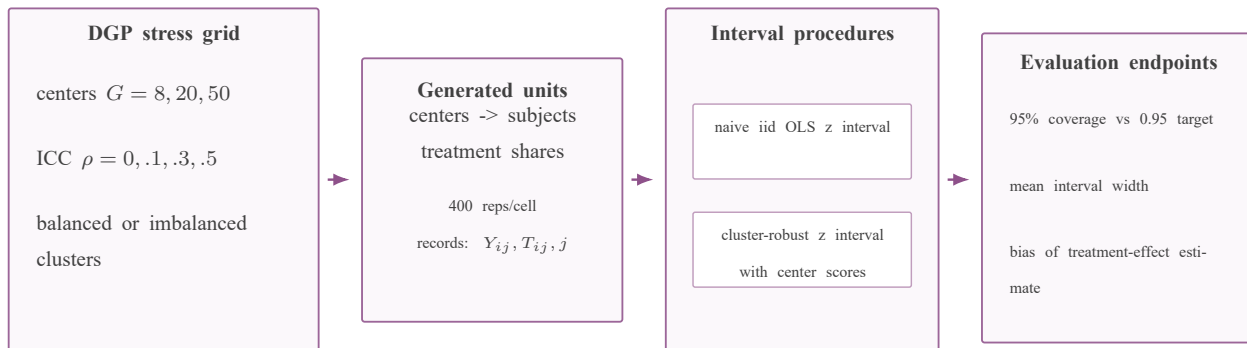
Cluster-robust intervals recover coverage as ICC increases



Red = naive iid; teal = cluster-robust. Nominal 0.95 and the small-center stress point remain visible.

Synthetic clustered-data example; intervals compared on coverage, width, and bias.

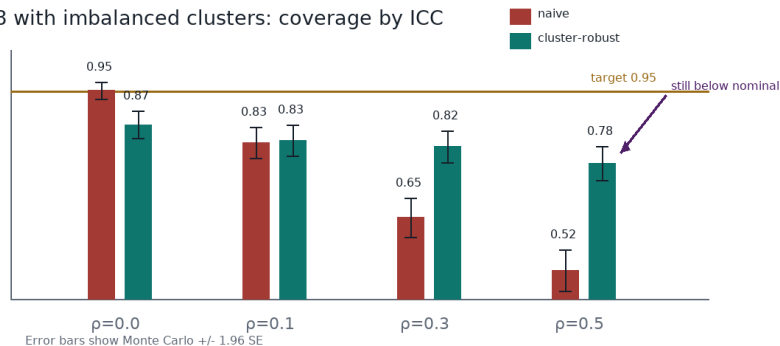
Simulation varies clustering before interval procedures are compared



The connector direction encodes data generation before interval estimation and coverage diagnostics.

Small-G settings remain anti-conservative after robustification

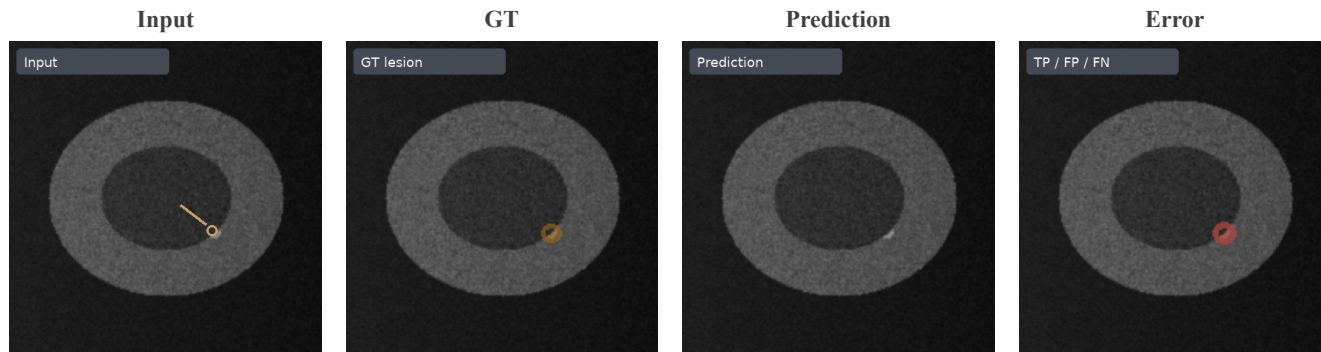
G=8 with imbalanced clusters: coverage by ICC



The failure region is adjacent to the diagnostic explanation instead of hidden in a note.

Negative evidence is completed; CR2 and wild cluster bootstrap remain the next test.

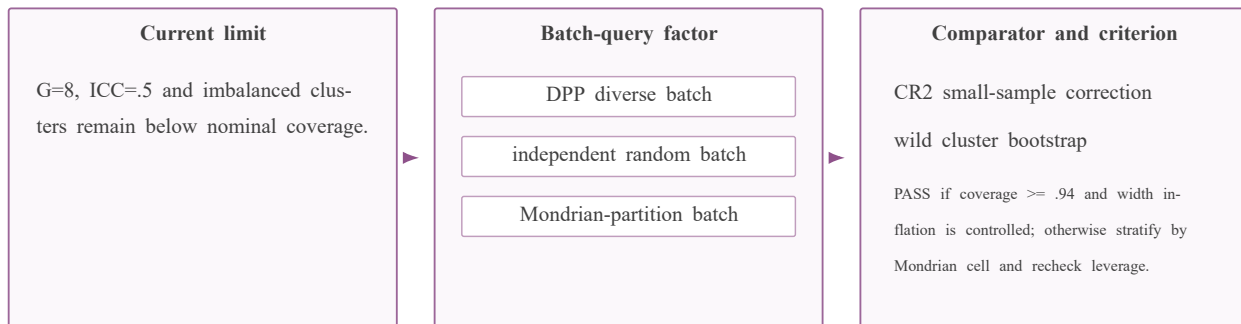
Same-case panels keep the segmentation error interpretable



**Error zoom: TP / FP / FN colors
remain bound to the same case**

All four panels come from the same case; the error panel binds the failure explanation.

Next experiment tests whether batch selection reduces fragile coverage



Coverage evidence determines which batch-query strategy to validate next; CR2 and wild bootstrap remain proposed comparators.